APC 523 PSET 3

Reyna Ayala

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1 Implicit and explicit analogues of common methods

(a) *Consider the two-stage fully implicit Runge-Kutta method. Perform the linear stability analysis of this scheme and analyze the properties of the stability region. Would you recommend using this fourth-order scheme in practice?*

Taking the following,

$$u^{n+1} = u^n e^{\Omega \Delta t}, \text{ with } u(t) = u_0 e^{\Omega t} \text{ being the solution of } \frac{du}{dt} = f(u)$$

then for two-stage fully implicit RK we can write

$$k_1 = f\left(u^n + \frac{\Delta t}{4} k_1 + \Delta t k_2 \left(\frac{1}{4} - \frac{\sqrt{3}}{6}\right)\right)$$

$$\therefore k_1 = \Omega u^n + \frac{\Omega \Delta t}{4} k_1 + \Omega \Delta t k_2 \left(\frac{1}{4} - \frac{\sqrt{3}}{6}\right) \tag{1}$$

and

$$k_2 = \Omega u^n + \Omega \Delta t k_1 \left(\frac{1}{4} + \frac{\sqrt{3}}{6}\right) + \frac{\Omega \Delta t}{4} k_2 \tag{2}$$

I solved for k_1 and k_2 from (1) and (2) using Mathematica:

```
In[•]:= eqns = {
          k1 == A*u + A*t*k1/4 + A*t*k2*(1/4 - Sqrt[3]/6),
          k2 == A*u + A*t*k1*(1/4 + Sqrt[3]/6) + A*t/4*k2
        };

      Solve[eqns, {k1, k2}]
```

Out[•]= $\left\{\left\{k1 \to -\frac{2\left(-6A + \sqrt{3}A^2 t\right)u}{12 - 6At + A^2 t^2},\ k2 \to \frac{2A\left(6 + \sqrt{3}At\right)u}{12 - 6At + A^2 t^2}\right\}\right\}$

Figure 1: Mathematica code to solve for k_1 and k_2 in implicit RK analysis.

Thus,

$$k_1 = -\frac{2u^n(-6\Omega + \sqrt{3}\Omega^2\Delta t)}{\Omega^2\Delta t^2 - 6\Omega\Delta t + 12},$$
$$k_2 = \frac{2u^n\Omega(6 + \sqrt{3}\Omega\Delta t)}{\Omega^2\Delta t^2 - 6\Omega\Delta t + 12}$$

Returning to the update equation,

$$\begin{aligned} u^{n+1} &= u^n + \Delta t\left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right) \\ &= u^n + \Delta t\left(-u^n\frac{-6\Omega + \sqrt{3}\Omega^2\Delta t}{\Omega^2\Delta t^2 - 6\Omega\Delta t + 12} + u^n\frac{6\Omega + \sqrt{3}\Omega^2\Delta t}{\Omega^2\Delta t^2 - 6\Omega\Delta t + 12}\right) \\ &= u^n\left(1 + \frac{12\Omega\Delta t}{\Omega^2\Delta t^2 - 6\Omega\Delta t + 12}\right) \end{aligned}$$

Therefore, the amplification factor is

$$\rho = 1 + \frac{12\Omega\Delta t}{\Omega^2\Delta t^2 - 6\Omega\Delta t + 12} \tag{3}$$

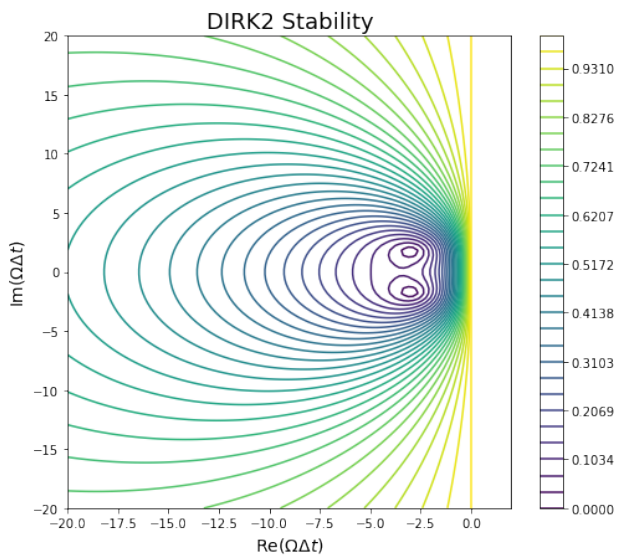


Figure 2: Amplification factor at varied $\Omega_r\Delta t$ and $\Omega_i\Delta t$, stable for $|\rho| < 1$.

I would recommend using the two-stage implicit Runge-Kutta method for stiff systems, which would otherwise require a very small timestep to remain stable for fully explicit methods. Being implicit, this method requires a nonlinear solve per stage. This method is L-stable, guaranteeing convergence to the equilibrium state. Though, the nonlinear solve will still add computational complexity and cost and is thus primarily worth the effort for stiff systems.

(b) *Explicit analogues of the Backward Differentiation Formulas have the general form*

$$u^{n+1} + \sum_{i=0}^{S} \alpha_i u^{n-1} = \beta f(u^{n+1})$$

Derive the coefficients α_i and β for both second-order and third-order methods. Perform the linear stability analysis and plot the stability region. Would you ever want to use these methods in practice?

I used [2] and [3] to inform my analysis approach to this problem, along with class lecture notes.

I note that $\beta = \Delta t$ and

$$\left.\frac{du}{dt}\right|_{t=t^{n+1}} = f(u^{n+1})$$

For second order ($S = 1$),

$$\begin{aligned} g(t) &= u^{n+1} \cdot \frac{(t-t^n)(t-t^{n-1})}{(t^{n+1}-t^n)(t^{n+1}-t^{n-1})} + u^n \cdot \frac{(t-t^{n+1})(t-t^{n-1})}{(t^n-t^{n+1})(t^n-t^{n-1})} + u^{n-1} \cdot \frac{(t-t^{n+1})(t-t^n)}{(t^{n-1}-t^{n+1})(t^{n-1}-t^n)} \\ &= u^{n+1} \cdot \frac{(t-t^n)(t-t^{n-1})}{2\Delta t^2} + u^n \cdot \frac{(t-t^{n+1})(t-t^{n-1})}{-2\Delta t^2} + u^{n-1} \cdot \frac{(t-t^{n+1})(t-t^n)}{2\Delta t^2} \end{aligned}$$

To find each coefficient, we evaluate the derivative of each numerator at $t = t^{n+1}$:

$$\left.\frac{d}{dt}\left[(t-t^n)(t-t^{n-1})\right]\right|_{t=t^{n+1}} = \left.2t - (t^n + t^{n-1})\right|_{t=t^{n+1}} = 3\Delta t$$

$$\left.\frac{d}{dt}\left[(t-t^{n+1})(t-t^{n-1})\right]\right|_{t=t^{n+1}} = 4\Delta t$$

$$\left.\frac{d}{dt}\left[(t-t^{n+1})(t-t^{n})\right]\right|_{t=t^{n+1}} = \Delta t$$

Thus,

$$g'(t) = \frac{1}{\Delta t}\left(\frac{3}{2}u^{n+1} - 2u^n + \frac{1}{2}u^{n-1}\right)$$

This leads to the update equation

$$u^{n+1} - \frac{4}{3}u^n + \frac{1}{3}u^{n-1} = \frac{2}{3}\Delta t f(u^{n+1})$$

Using $u^{n+1} = \rho u^n = \rho^2 u^{n-1}$ gives

$$\rho^2 u^{n-1} - \frac{4}{3}\rho u^{n-1} + \frac{1}{3}u^{n-1} = \frac{2}{3}\Delta t \Omega \rho^2 u^{n-1}$$

$$(3 - 2\Omega\Delta t)\rho^2 - 4\rho + 1 = 0$$

$$\rho = \frac{4 \pm \sqrt{4 + 8\Omega\Delta t}}{6 - 4\Omega\Delta t} \tag{4}$$

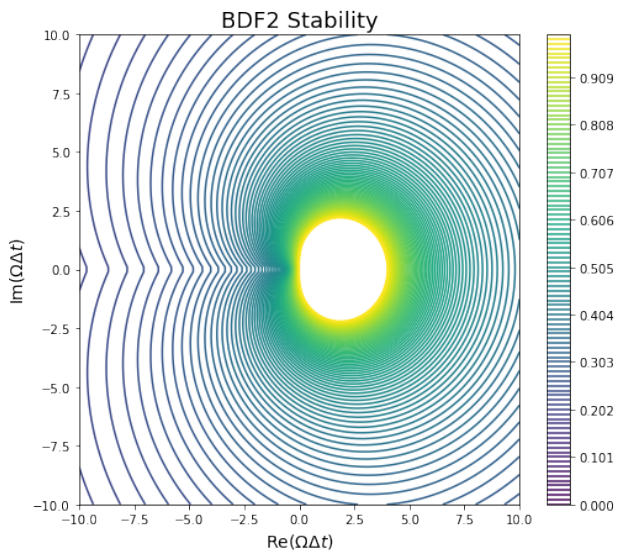


Figure 3: Amplification factor at varied $\Omega_r\Delta t$ and $\Omega_i\Delta t$, stable for $|\rho| < 1$.

For third-order ($S = 2$),

$$\begin{aligned} g(t) =& u^{n+1} \cdot \frac{(t-t^n)(t-t^{n-1})(t-t^{n-2})}{6\Delta t^3} + u^n \cdot \frac{(t-t^{n+1})(t-t^{n-1})(t-t^{n-2})}{-2\Delta t^3} \\ &+ u^{n-1} \cdot \frac{(t-t^{n+1})(t-t^n)(t-t^{n-2})}{2\Delta t^3} + u^{n-2} \cdot \frac{(t-t^{n+1})(t-t^n)(t-t^{n-1})}{-6\Delta t^3} \end{aligned}$$

Following the same procedure as the second-order BDF analysis,

$$\begin{aligned} \frac{d}{dt}\left[(t-t^n)(t-t^{n-1})(t-t^{n-2})\right]\Big|_{t=t^{n+1}} &= (t-t^n)(t-t^{n-1}) + (t-t^n)(t-t^{n-2}) + (t-t^{n-1})(t-t^{n-2})\Big|_{t=t^{n+1}} \\ &= 11\Delta t^2 \end{aligned}$$

$$\frac{d}{dt}\left[(t-t^{n+1})(t-t^{n-1})(t-t^{n-2})\right]\Big|_{t=t^{n+1}} = 6\Delta t^2$$

$$\frac{d}{dt}\left[(t-t^{n+1})(t-t^n)(t-t^{n-2})\right]\Big|_{t=t^{n+1}} = 3\Delta t^2$$

$$\frac{d}{dt}\left[(t-t^{n+1})(t-t^{n})(t-t^{n-1})\right]\Big|_{t=t^{n+1}} = 2\Delta t^2$$

Thus,

$$g'(t) = \frac{1}{\Delta t}\left(\frac{11}{6}u^{n+1} - 3u^n + \frac{3}{2}u^{n-1} - \frac{1}{3}u^{n-2}\right)$$

making the update equation

$$\frac{11}{6}u^{n+1} - 3u^n + \frac{3}{2}u^{n-1} - \frac{1}{3}u^{n-2} = \Omega\Delta t u^{n+1}$$

Thus,

$$\frac{11}{6}\rho^3 u^{n-2} - 3\rho^2 u^{n-2} + \frac{3}{2}\rho u^{n-2} - \frac{1}{3}u^{n-2} = \Omega\Delta t\rho^3 u^{n-2}$$

This leads to the cubic equation

$$\left(1 - \frac{6}{11}\Omega\Delta t\right)\rho^3 - \frac{18}{11}\rho^2 + \frac{9}{11}\rho - \frac{2}{11} = 0 \tag{5}$$

with the stability region plotted in 4

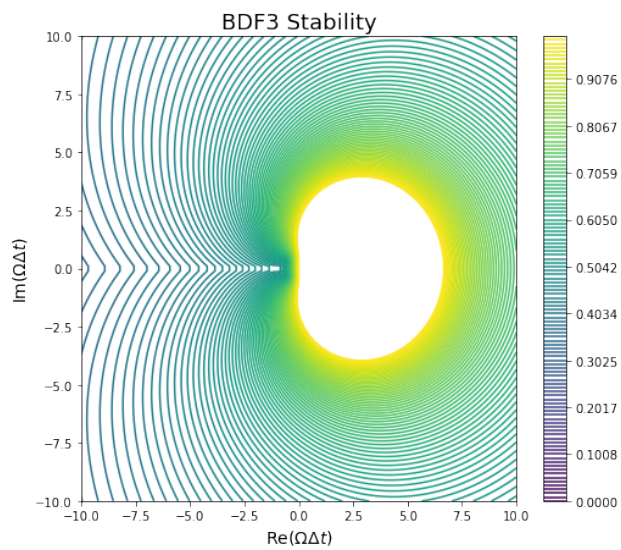


Figure 4: Amplification factor at varied $\Omega_r\Delta t$ and $\Omega_i\Delta t$, stable for $|\rho| < 1$.

BDF methods are much cheaper then implicit RK methods, only requiring one nonlinear solve per step. These methods are also L-stable and well-suited to solving stiff systems. However, BDF methods require use of another method to start.

2 Harmonically forced undamped oscillator

The displacement $x(t)$ of a forced undamped harmonic oscillator is described by the initial value problem

$$m\ddot{x} + kx = F\cos(\omega_F t)$$

with initial position $x(0) = x_0$ and velocity $v(0) = \dot{x}(0) = v_0$. Above, m is the mass of the oscillator, k is the spring constant and F and ω_F are the magnitude and frequency of the forcing. If we define the natural frequency of the oscillator $\omega = \sqrt{k/m}$, the governing equations may be rewritten as:

$$\frac{d^2x}{dt^2} + \omega^2 x = \frac{F}{m}\cos(\omega_F t)$$

For initial data (x_0, v_0) with parameters $\omega = 5$, $F/m = 1$, and $\omega_F = 0.1$, integrate the ODE over $t \in [0, 100]$ using each of the following methods:

(a) *Forward Euler*

(b) *Symplectic Euler*

(c) *Fourth-order Runge-Kutta (RK4)*

How do you expect the responses of this system to look? In particular, discuss the properties of the solution associated with the fast and slow frequencies. For each numerical integration method, plot the response of the system ($x(t)$ vs. t), its phase portrait ($v(t)$ vs. $x(t)$) and the L^2 error $||e(t)||_{L^2}$ of the numerical solution compared to the analytical solution at each point in time. Discuss the strengths of each approach. Which method gives the highest degree of accuracy for a given amount of CPU time? Do any of the methods conserve energy?

The oscillator is being forced at $\omega_F \ll \omega$ where ω is the natural frequency of the system. Thus, I expect the system to exhibit fast oscillations with ω_F overlayed with a slow oscillation with ω.

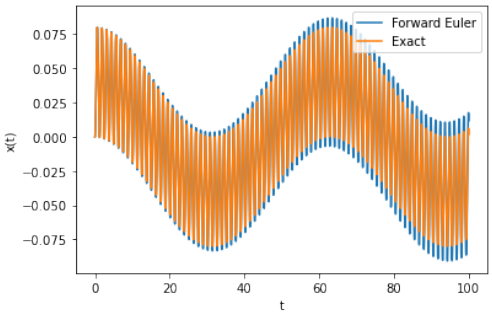
The response plot, phase portrait, and L^2 error for each method are shown in Figures 5, 6, and 7 for forward Euler, symplectic Euler, and RK4, respectively.

Forward Euler is conceptually the simplest time integration scheme applied, however a very small timestep is required to maintain numerical stability ($\Delta t = 0.0002$ is used). Even with a stable solution, there are significant amplitude and phase errors accumulated over the integration time.

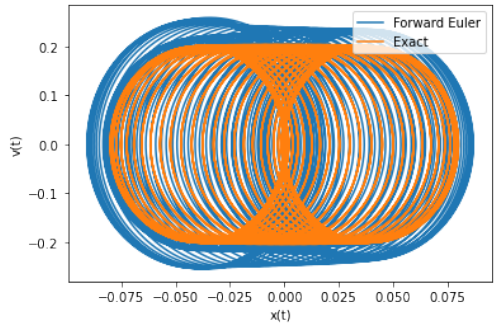
Symplectic Euler conserves a first order approximation of the true energy ("shadow Hamiltonian") and thus strictly preserves the amplitude. At first order, though, phase error is significant. The stability characteristics are much better than forward Euler, as a much smaller timestep can be used while maintaining stability ($\Delta t = 0.001$ is used here).

Fourth-order Runge-Kutta adds some complexity relative to forward Euler or symplectic Euler but achieves higher accuracy than either and improves stability characteristics compared to forward Euler ($\Delta t = 0.001$ is used here). RK4 does not conserve energy as symplectic Euler does, though this may not be important for the demands of the problem.

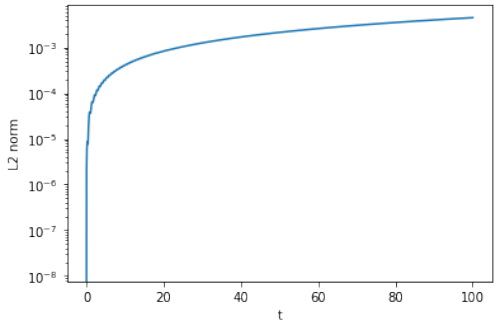
For this time interval of interest $t \in [0, 100]s$, RK4 just outperforms symplectic Euler when considering computation time versus L^2 accrued error. In Figure 8, we observe that the accrued L^2 error differs between symplectic Euler and RK4 by almost two orders of magnitude, whereas computation time differs by only one order of magnitude.



(a) FE response

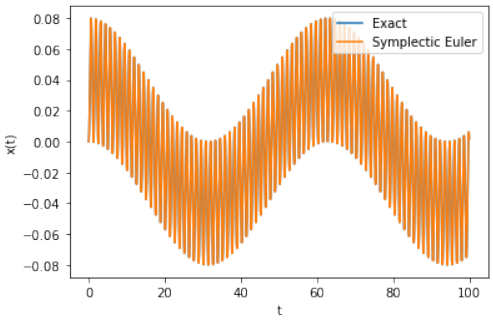


(b) FE phase portrait

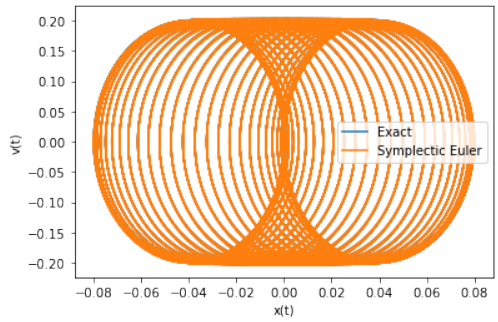


(c) FE L^2 error

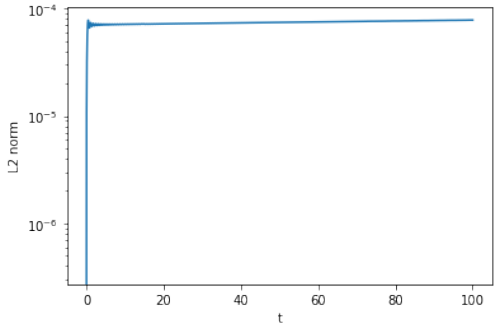
Figure 5: Forward Euler method applied to forced harmonic oscillator



(a) SE response

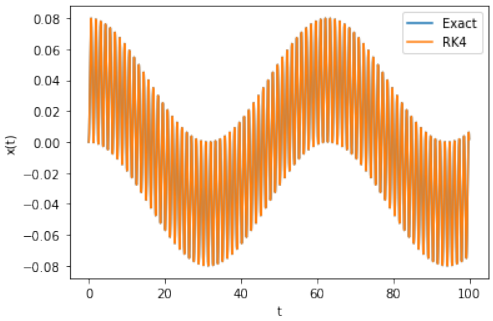


(b) SE phase portrait

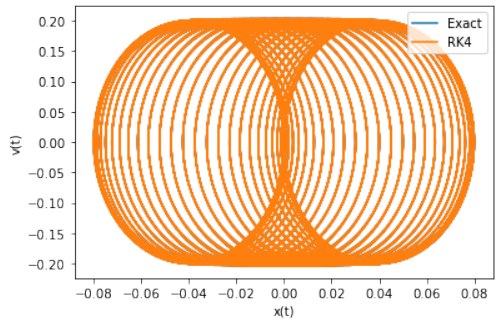


(c) SE L^2 error

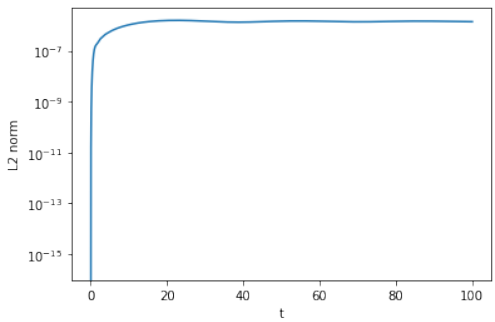
Figure 6: Symplectic Euler method applied to forced harmonic oscillator



(a) RK4 response



(b) RK4 phase portrait



(c) RK4 L^2 error

Figure 7: RK4 method applied to forced harmonic oscillator

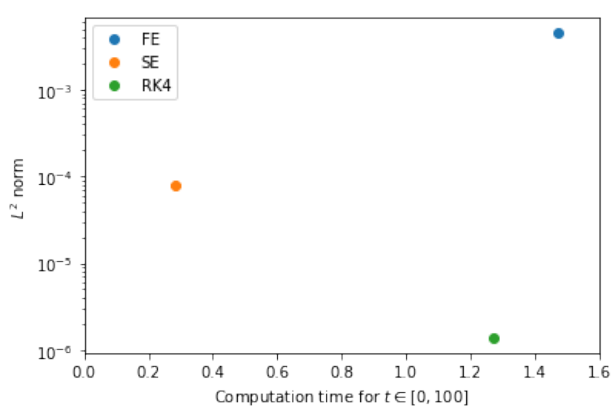


Figure 8: Computation time versus accrued error for the time interval of interest.

3 Conserved energy for the Verlet scheme

The undamped harmonic oscillator without forcing is described by the following ODE

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

with initial position $x(0) = x_0$ and velocity $v(0) = \dot{x}(0) = v_0$. We defined the modified energy:

$$\hat{H} = \frac{1}{2}v^2 + \frac{1}{2}\omega^2 x^2 \left(1 - \frac{\omega^2(\Delta t)^2}{4}\right)$$

Show analytically that $\hat{H}$ is strictly conserved by the second-order Verlet scheme for the simple harmonic oscillator.

Applying the Verlet scheme gives

$$v^{n+1/2} = v^n - \omega^2 x^n \frac{\Delta t}{2}$$

$$x^{n+1} = x^n + \Delta t v^n - \frac{1}{2}\Delta t^2 \omega^2 x^n$$

$$\begin{aligned} v^{n+1} &= v^n - \omega^2 x^n \frac{\Delta t}{2} - \omega^2 \frac{\Delta t}{2}(x^n + \Delta t v^n - \frac{1}{2}\Delta t^2 \omega^2 x^n) \\ &= v^n - \omega^2 \Delta t x^n - \frac{1}{2}\omega^2 \Delta t^2 v^n + \frac{1}{4}\omega^4 \Delta t^3 x^n \end{aligned}$$

Evaluating the first term of the modified energy equation,

$$\begin{aligned} \frac{1}{2}(v^{n+1})^2 = \frac{1}{2}(v^n)^2 + \frac{1}{2}\omega^4 \Delta t^2 (x^n)^2 + \frac{1}{8}\omega^4 \Delta t^4 (v^n)^2 + \frac{1}{32}\omega^8 \Delta t^6 (x^n)^2 - \omega^2 \Delta t x^n v^n - \frac{1}{2}\omega^2 \Delta t^2 (v^n)^2 \\ + \frac{1}{4}\omega^4 \Delta t^3 x^n v^n + \frac{1}{2}\omega^4 \Delta t^3 x^n v^n - \frac{1}{4}\omega^6 \Delta t^4 (x^n)^2 - \frac{1}{8}\omega^6 \Delta t^5 v^n x^n \end{aligned}$$

Evaluating half the square of x^{n+1},

$$\frac{1}{2}(x^{n+1})^2 = \frac{1}{2}(x^n)^2 + \frac{1}{2}\Delta t^2 (v^n)^2 + \frac{1}{8}\Delta t^4 \omega^4 (x^n)^2 + \Delta t v^n x^n - \frac{1}{2}\Delta t^2 \omega^2 (x^n)^2 - \frac{1}{2}\Delta t^3 \omega^2 x^n v^n$$

and so,

$$\begin{aligned} \frac{1}{2}(x^{n+1})^2 \left[\omega^2 - \frac{1}{4}\omega^4 \Delta t^2\right] = \frac{1}{2}\omega^2 (x^n)^2 + \frac{1}{2}\omega^2 \Delta t^2 (v^n)^2 + \frac{1}{8}\omega^6 \Delta t^4 (x^n)^2 + \omega^2 \Delta t v^n x^n \\ - \frac{1}{2}\omega^4 \Delta t^2 (x^n)^2 - \frac{1}{2}\omega^4 \Delta t^3 x^n v^n - \frac{1}{8}\omega^4 \Delta t^2 (x^n)^2 - \frac{1}{8}\omega^4 \Delta t^4 (v^n)^2 \\ - \frac{1}{32}\omega^8 \Delta t^6 (x^n)^2 - \frac{1}{4}\omega^4 \Delta t^3 v^n x^n + \frac{1}{8}\omega^6 \Delta t^4 (x^n)^2 + \frac{1}{8}\omega^6 \Delta t^5 x^n v^n \end{aligned}$$

Notice that most terms cancel when the two expressions are added, leaving

$$\frac{1}{2}(v^{n+1})^2 + \frac{1}{2}(x^{n+1})^2 \left(\omega^2 - \frac{1}{4}\omega^4 \Delta t^2\right) = \frac{1}{2}(v^n)^2 + \frac{1}{2}(x^n)^2 \left(\omega^2 - \frac{1}{4}\omega^4 \Delta t^2\right)$$

Therefore, the modified energy $\hat{H}$ is conserved by the Verlet scheme.

4 Stiff systems in chemical kinetics

Stiff systems of ordinary differential equations are commonly encountered in chemical kinetics. Consider the following set of chemical reactions:

$$\frac{d}{dt}\begin{bmatrix} X_N \\ X_O \\ X_{N_2} \\ X_{O_2} \\ X_{NO} \end{bmatrix} = \begin{bmatrix} k_{+1}X_{N_2}X_O - k_{-1}X_{NO}X_N - k_{+2}X_N X_{O_2} \\ -k_{+1}X_{N_2}X_O + k_{-1}X_{NO}X_N + k_{+2}X_N X_{O_2} \\ -k_{+1}X_{N_2}X_O + k_{-1}X_{NO}X_N \\ -k{+}2X_N X_{O_2} \\ k_{+1}X_{N_2}X_O - k_{-1}X_{NO}X_N + k_{+2}X_N X_{O_2} \end{bmatrix}$$

The reaction rate coefficients are

$$k_{+1} = 2 \times 10^3$$
$$k_{-1} = 3 \times 10^{-12}$$
$$k_{+2} = 2 \times 10^1$$

(a) *Given the initial conditions*

$$\begin{bmatrix} X_N \\ X_O \\ X_{N_2} \\ X_{O_2} \\ X_{NO} \end{bmatrix} = \begin{bmatrix} 0.01 \\ 0.01 \\ 0.75 \\ 0.23 \\ 0.00 \end{bmatrix}$$

compute the eigenvalues of the Jacobian of the system at the initial time. Is the system stiff?

The eigenvalues of the Jacobian are:

$$\{-1.535 \times 10^3, 1.011 \times 10^1, -2.012 \times 10^{-1}, 9.291 \times 10^{-15}, -1.388 \times 10^{-30}\}$$

Thus,

$$\frac{|\lambda|_{max}}{|\lambda|_{min}} \sim 10^{33} \gg 1$$

and the system is very stiff.

(b) *Solve the evolution of the mole fractions with the provided initial conditions until $t = 40s$ using:*

i. *Fourth-order Runge-Kutta (RK4)*
ii. *Second-order Diagonally Implicit Runge-Kutta (DIRK2)*
iii. *Second-order Implicit Backward Differentiation Formulas (BDF2)*
iv. (ii.) *with adaptive time-stepping*
v. (iii.) *with adaptive time-stepping*

Plot the evolution of the system in time (log-log). How do the three methods compare in terms of required computational cost? Would you choose a BDF method over another implicit method like DIRK at the same order?

Note that I used [1] accompanied by lecture notes to inform my understanding of the Newton-Raphson method for a nonlinear solve.

Method	Runtime (s)
RK4	3.45
DIRK2	6.14
BDF2	3.28
adaptive DIRK2	0.0198
adaptive BDF2	0.0145

Table 1: Computational cost comparison of implemented methods

The implemented methods have run times summarized in Table 1.

Of the fixed timestep methods, BDF2 is fastest (though comparable to RK4). Since BDF2 is also implicit, it has superior stability characteristics to RK4 but can be more complex to implement given the nonlinear solve. Considering all implemented methods, the adaptive BDF2 had the shortest run time. If looking for computational efficiency, I would likely choose BDF over other implicit methods. However, if I wanted higher accuracy, higher-order BDF methods have poor stability characteristics, so I would likely sacrifice computational efficiency for accuracy by choosing a different method.

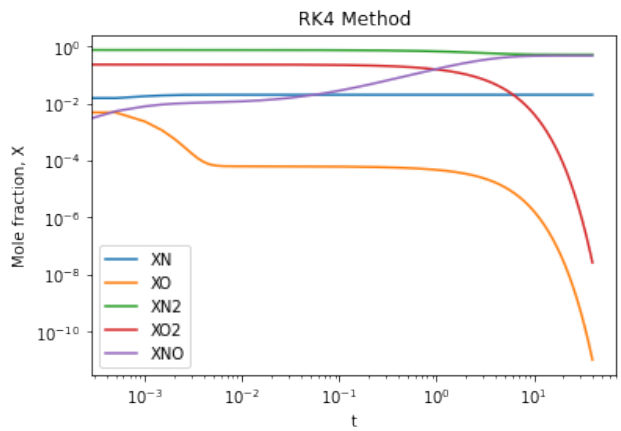


Figure 9: RK4 time integration for molar fractions

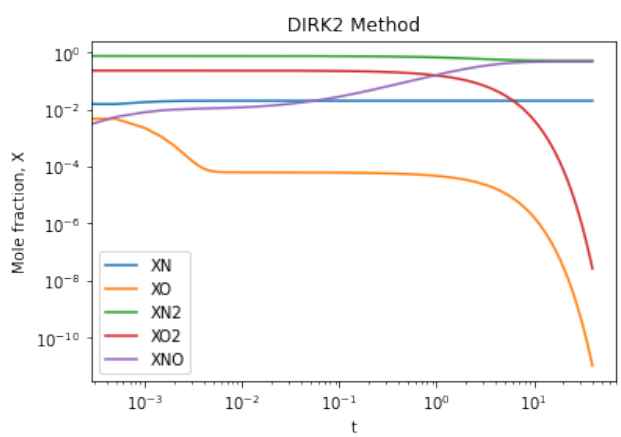


Figure 10: DIRK2 fixed-Δt time integration for molar fractions

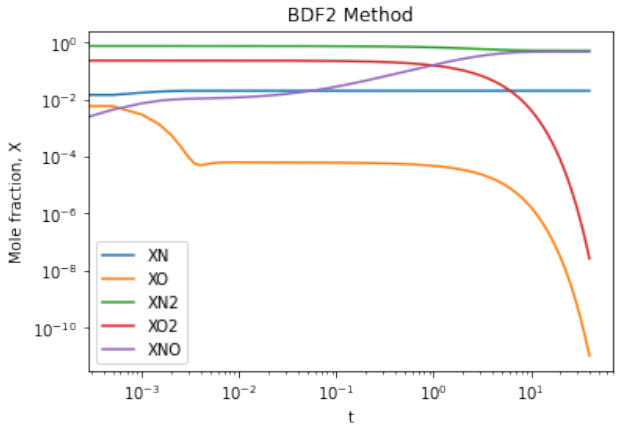


Figure 11: BDF fixed-Δt time integration for molar fractions

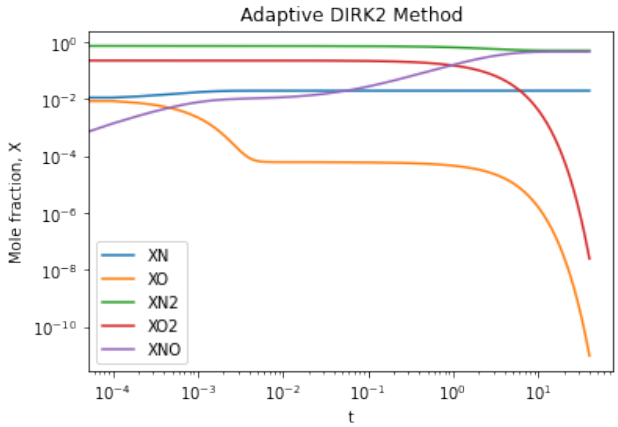


Figure 12: DIRK2 variable-Δt time integration for molar fractions

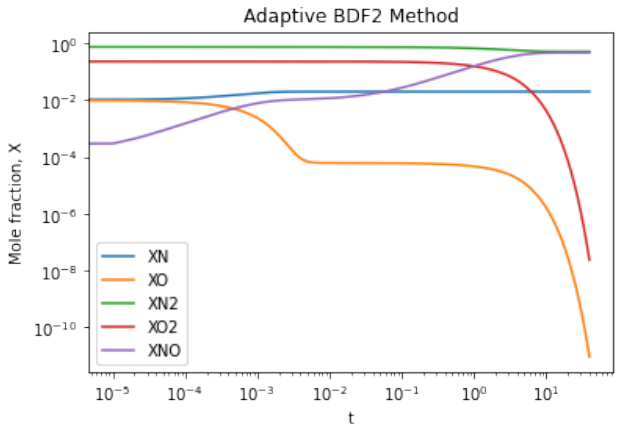


Figure 13: BDF2 variable-Δt time integration for molar fractions

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